Companions hidden in public HARPS velocities: four stellar companions with no published orbit

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Abstract

HARPS-RVBank holds 212 562 public velocities of 2 912 stars, and Gaia DR3 measures the same stars' motions on the sky. Searched together, they reveal companions that no catalogue holds. Every star whose velocities vary as a companion's would is set against SIMBAD, the papers it lists, Gaia DR3's two-body orbits and the NASA Exoplanet Archive, and the rest are fitted with one Keplerian orbit, weighed where the orbit outlasts the velocities against the Hipparcos-Gaia proper-motion anomaly, Gaia's acceleration and trend solutions and the range of its own epoch velocities. Four stellar companions follow with no published orbit: a $0.41\,M_\odot$ star seen nearly face on in a 465 d orbit of the F5 dwarf HD 7399, whose velocities alone give a brown dwarf's minimum mass; a $0.21\,M_\odot$ star in a 26 yr orbit of the K7 dwarf GJ 4001; a companion of at least $0.30\,M_\odot$ in an eccentric 344 d orbit of the F7 dwarf HD 77462; and a star of $1.15 \pm 0.07\,M_\odot$ in a $27.2^{+2.0}_{-1.5}$ yr orbit of the $2.92\,M_\odot$ G8 giant HD 20037, with an eccentricity of 0.45 ± 0.02 and an inclination of $98.4 \pm 1.1°$, from the giant's HARPS velocities, Gaia DR3's velocity-trend and acceleration solutions and its Hipparcos-Gaia proper-motion anomaly fitted together. The giant's spectra hold G-type lines where that companion's track predicts them, at 1.7 per cent of a G0 dwarf's line depth and 5.4σ against permuted tracks.

Keywords: Techniques: radial velocities, Astrometry, Binaries: spectroscopic, Stars: individual: HD 20037

1. Introduction

Radial-velocity surveys have watched thousands of nearby stars for two decades, and their archives hold more than their programmes published. HARPS-RVBank [1] recomputed every public HARPS velocity with SERVAL [2], corrected the instrument's nightly zero points, and attached each spectrum's activity indicators, so that a star's whole record can be read at once. Gaia DR3 adds two independent views of the same stars: its own spectroscopic velocities and their variability, with orbital and trend solutions for some, and astrometry whose acceleration and proper-motion anomaly against Hipparcos [3] bound a companion's mass and inclination [4].

This paper reads the two together for companions that no catalogue holds. Section 2 describes the data, Section 3 the stellar and substellar companions and Section 4 the companion of the giant HD 20037.

2. Data

The velocities are HARPS-RVBank's SERVAL velocities with their nightly zero points corrected, 2 912 stars and 212 562 velocities, with the chromatic index, differential line width, Hα

and Na D indices and the pipeline's line width and bisector of each spectrum. Each star is matched to Gaia DR3 [4] as the brightest source within 30 arcsec of its J2000 position whose G lies within 2.5 mag of its V, which follows stars of high proper motion to Gaia's epoch, and to the Hipparcos-Gaia proper-motion anomaly of ref. [3] through its Hipparcos number, with that catalogue's mass of the star. The public HARPS spectra of the stars discussed here come from the ESO Science Archive, and their ultraviolet and X-ray fluxes from GALEX [5], eROSITA's first survey [6], ROSAT's 2RXS [7] and 4XMM. Every analysis is that of orbitlight (https://huggingface.co/datasets/phanerozoic/orbitlight), whose tables give each star's decision and each orbit's posterior draws.

3. Stellar and substellar companions

A star enters the search when its nightly velocities vary as a companion's would, with a reduced χ^2 above 25 and a scatter above $30\,\mathrm{m\,s^{-1}}$ and above five times the median error on four nights or more, and when it is of spectral type F, G, K or M and luminosity class III or fainter; the signals of the planet search with a minimum mass above $13\,\mathrm{M_J}$ or a semi-amplitude of $100\,\mathrm{m\,s^{-1}}$ or more join them. Of 341 stars that vary, 273 are known: SIMBAD lists each as a binary or as a host of planets or brown dwarfs, a paper on a binary, a companion or an orbit covering at most 500 objects names it, Gaia DR3 holds a two-body orbit for it, or the NASA Exoplanet Archive lists planets around it. Another 55 have fewer than ten nights. The remaining 13, with 58 stars whose planet-search signals are that heavy, are analysed. An eccentric periodogram gives each star's best orbit, its false-alarm level from velocities permuted within each series, and the orbit is sampled where the velocities cover two cycles. Where they cover fewer, one Keplerian orbit of the star about the barycentre, the companion taken dark and no heavier than the star, is fitted to the velocities, the Hipparcos-Gaia proper-motion anomaly [3], Gaia DR3's acceleration and velocity-trend solutions and, where Gaia's spectrometer saw the velocity vary, the range of its epoch velocities less $2.5\,\mathrm{km\,s^{-1}}$ as a lower bound on the orbit's range over Gaia's window. A profile over period, the best orbit in each of 120 bands of log period found by differential evolution, gives the periods the data allow, those within 2 of the best in log posterior, and the posterior is sampled by an ensemble of differential-evolution moves. A companion is accepted where a second reduction of the archived spectra, by cross-correlation, follows the velocities with a correlation of 0.9 or more, and either the velocities cover two cycles with a false-alarm level below 10^{-3} and no other period within 10 in χ^2, or the astrometry is significant (an anomaly of 5σ, or a Gaia acceleration or trend solution) and the allowed periods span less than a factor of two clear of the prior's limits. SIMBAD's references and the papers they cite give each star's literature.

Three companions have no published orbit (Table 1). HD 7399, an F5 dwarf, moves on a 465 d orbit at only $0.52\,\mathrm{km\,s^{-1}}$, a minimum mass of $0.023\,\mathrm{M_\odot}$ in the brown-dwarf desert, but its proper-motion anomaly of 15σ, Gaia's acceleration solution (Acceleration9) and its RUWE of 1.87 move it far more on the sky than that mass could: the orbit is seen within 4° of face on and the companion is a $0.41\,\mathrm{M_\odot}$ star. GJ 4001 (MCC 791), a K7 dwarf, has a companion of at least $0.041\,\mathrm{M_\odot}$ on an eccentric 26 yr orbit that its six-year series covers in part, and its anomaly of 64σ and RUWE of 9.2 put the orbit 13° from face on and the companion at $0.21\,\mathrm{M_\odot}$, an M dwarf. HD 77462, an F7 dwarf, has 11 HARPS nights over 472 d, and Gaia's epoch velocities span $20.7\,\mathrm{km\,s^{-1}}$; together they give an orbit of 344 d with an eccentricity of 0.68 and a companion of at least $0.30\,\mathrm{M_\odot}$. Fitted as dark, its anomaly of 5.8σ and RUWE of 14.0 make it $1.23\,\mathrm{M_\odot}$ at $i = 161°$, which a main-sequence star of

Table 1: Companions with no published orbit: the nights of velocities, the periods the data allow and the orbit's period, semi-amplitude and eccentricity with their 16th to 84th percentiles, the primary's mass m_1 [3], the companion's minimum mass and its mass from the astrometry, and Gaia's RUWE and the Hipparcos-Gaia proper-motion anomaly's signal to noise.

Star	Type	Nights	P (d), allowed	K ($\mathrm{km\,s^{-1}}$)	e	m_1 ($\mathrm{M_\odot}$)	$m_2 \sin i$ ($\mathrm{M_\odot}$)	m_2 ($\mathrm{M_\odot}$)	RUWE, PMa S/N
HD 7399	F5 V	22	465 (460–473), 441–478	0.523 (0.516–0.533)	0.07 (0.05–0.12)	1.30	0.023	0.41 (0.38–0.44)	1.87, 15.1
GJ 4001	K7 V	26	9380 (8660–10210), 7540–11350	0.833 (0.815–0.852)	0.76 (0.75–0.78)	0.63	0.041	0.21 (0.19–0.24)	9.24, 64.3
HD 77462	F7 V	11	344.0 (343.8–344.2), 330–359	8.86 (8.41–9.61)	0.68 (0.66–0.70)	1.30	0.30 (0.28–0.31)	1.23 (1.13–1.37)	14.0, 5.8

Table 2: HD 20037's orbit from its HARPS velocities and Gaia's and Hipparcos's astrometry together: medians with 16th to 84th percentiles of the sampled posterior.

Period (d)	9 953 (9 384–10 666)
Eccentricity	0.45 (0.43–0.47)
Giant's semi-amplitude K_1 ($\mathrm{km\,s^{-1}}$)	4.99 (4.94–5.04)
Inclination (deg)	98.4 (97.4–99.6)
Semi-major axis (AU)	14.4 (13.7–15.1)
Companion's mass ($\mathrm{M_\odot}$)	1.15 (1.08–1.21)

that mass cannot be: it would give nearly as much light as the primary and move the photocentre little, so the companion is compact or the astrometric excess has a further source.

4. The companion of the giant HD 20037

HD 20037 is a G8 giant of $2.92 \pm 0.15\,\mathrm{M_\odot}$ and $10.3\,\mathrm{R_\odot}$ [3] at 172 pc. Its 13 HARPS velocities fall smoothly by $3.3\,\mathrm{km\,s^{-1}}$ from 2006 to 2011 with nights a day apart agreeing to a few metres per second. Gaia DR3 holds three more views of the same orbit. Its 27 velocities over 2014 to 2017 vary beyond their errors and are fitted by a second-degree trend (SecondDegreeTrendSB1) rising at $0.0071 \pm 0.0036\,\mathrm{km\,s^{-1}}$ per day at 2016.0; its astrometry carries an acceleration solution (Acceleration7) at a significance of 31.9, -1.468 ± 0.046 and $-0.705 \pm 0.047\,\mathrm{mas\,yr^{-2}}$ in right ascension and declination, with the proper motion at 2016.0; and its mean motion between Hipparcos and Gaia differs from that proper motion at 114σ [3].

One Keplerian orbit of the giant about the barycentre, the companion dark, is fitted to all of it at once. The HARPS velocities are read directly with a zero point and a jitter. Gaia's velocity derivatives are those of a quadratic fitted to the model's velocities at 27 times across its 968-day velocity window; the proper motion and acceleration at 2016.0 are those of a quadratic fitted to the photocentre over Gaia's astrometric window; and the long-term mean motion is the photocentre's displacement from its Hipparcos position, a line fitted over Hipparcos's window, to its Gaia position over the 24.75 yr between them. The parameters are the period, the time of periastron, $\sqrt{e}\cos\omega$, $\sqrt{e}\sin\omega$, the companion's mass as a fraction of the giant's, $\cos i$, the node, the HARPS zero point, the jitter and the giant's mass with a prior of 10 per cent; Gaia's 27 velocities span $7.7\,\mathrm{km\,s^{-1}}$, and less $2.5\,\mathrm{km\,s^{-1}}$ for their noise that range bounds the orbit's over Gaia's window from below. A profile over period located the posterior, which allows periods from 8 830 to 12 190 d, and an ensemble of differential-evolution moves sampled it (Table 2). The companion weighs $1.15 \pm 0.07\,\mathrm{M_\odot}$ on an orbit of 27.2 yr seen nearly edge on; periastron passed in 2013, between the HARPS and Gaia series.

A main-sequence star of $1.15\,\mathrm{M_\odot}$, an F8 or G0 dwarf, would give 3 to 4 per cent of the system's

V-band light. With the orbit known, the companion's lines can be sought along their own track: in each spectrum it moves against the giant at $-(1 + m_1/m_2)$ times the giant's velocity about the systemic one, here +8.1 to +20.0 km s^{-1}. The 13 spectra are placed in the giant's frame on a grid uniform in velocity, divided by their median, and cross-correlated with the line depths of the G0 dwarf ζ Tuc, each wavelength weighted by the spectra's scatter smoothed over 501 km s^{-1}, and the sky's pattern is taken out of every row in its own frame. Each row is then fitted as a pattern common to every row plus α times the G0 lines' own cross-correlation at the companion's velocity, the two by weighted least squares together, so that α is the depth of the companion's lines as a fraction of a G0 dwarf's and the part of the trail the median spectrum holds is fitted with it. Along the track $\alpha = 0.017$, at 5.4σ against 400 permutations of the track among the spectra, and the G0 spectrum laid into the spectra along the track at 1, 2 and 4 per cent of the light raises α by 0.010, 0.020 and 0.040. Along the track mirrored about the giant's velocity α is 0.002, and along the telluric lines' own track 0.003. The spectra see the companion only between +8 and +20 km s^{-1}, so a component at a fixed velocity near +12 km s^{-1} from the giant fits them as well as one on the orbit; Gaia DR3 holds no other source within 20 arcsec and finds no second peak in the star's images, which would leave such a component within about 0.2 arcsec of the giant. GALEX measures HD 20037 at FUV 19.76 ± 0.11 and NUV 14.24 ± 0.01, a G8 giant's colours with no hot component, and eROSITA records a soft source 1.8 arcsec from the star, 17 counts mostly between 0.5 and 1 keV, 2.0×10^{29} erg s^{-1} in the 0.2 to 2.3 keV band, as a giant's corona or an active dwarf gives.

The companion is therefore most likely a main-sequence star near 1.15 $M_\odot$, whose lines, at 1.7 per cent of a G0 dwarf's depth against the 3 to 4 per cent of the light such a star gives, are shallower than a slowly rotating G0 dwarf's, as rotation or a slightly earlier type makes them. The orbit's fit takes the companion as dark; at 1.7 per cent of the light the photocentre moves about 6 per cent less than the giant, which the fit reads as a slightly lighter companion.

5. Conclusion

Public HARPS velocities read beside Gaia DR3's astrometry and velocities give four stellar companions with no published orbit: a brown-dwarf-desert minimum mass that the astrometry turns into a 0.41 $M_\odot$ star seen face on around HD 7399, an M dwarf on a 26-year orbit of GJ 4001, a companion of HD 77462 heavier than 0.30 $M_\odot$ whose astrometric mass no main-sequence star could have, and a companion of 1.15 ± 0.07 $M_\odot$ on a 27-year orbit of the giant HD 20037, whose lines appear in the giant's spectra along its predicted track at 1.7 per cent of a G0 dwarf's depth.

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